

CIS 635 Knowledge Discovery & Data Mining

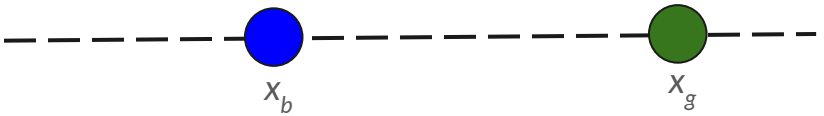
Introduction to Distance Metric + *k-NN*

Outline

- Proximity vs Distance Metric
- k-NN, our first ML model

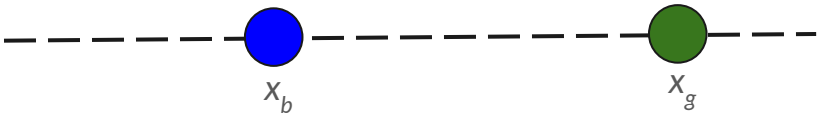
Distance (or Proximity) metric

- Let's we are give two data points: one, the **blue**, and the other is the **green** circle.



Distance (or Proximity) metric

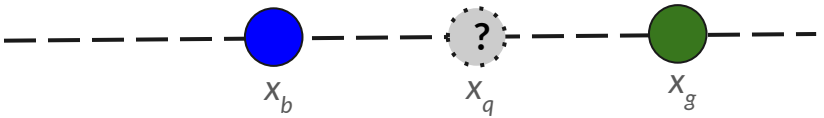
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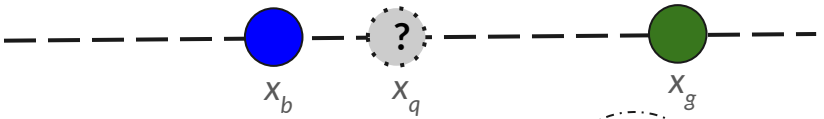


Which color do you think ? *is most likely?*



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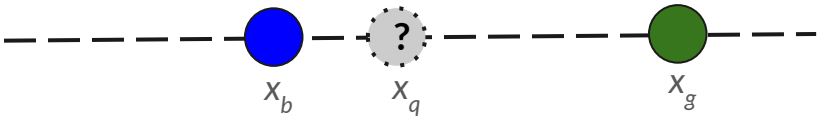


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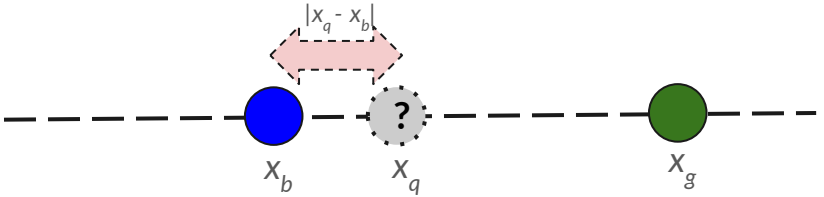


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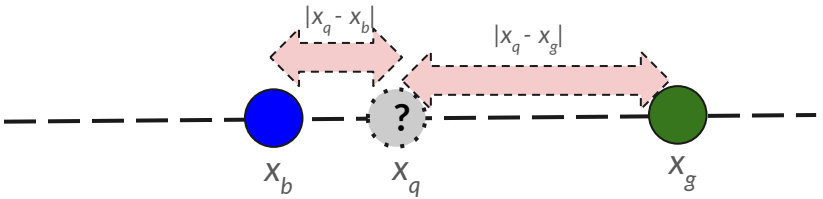


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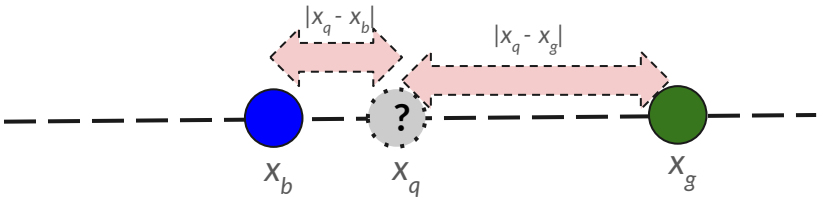


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Decision Rule: $|x_q - x_b| < |x_q - x_g|$



We learned about 3 Distance Metrics

Distance Ranges:

- **L1/Manhattan Distance**: [0 - ∞]
- **L2/Euclidean Distance**: [0 - ∞]
- **Cosine Distance**: [0 - 2]

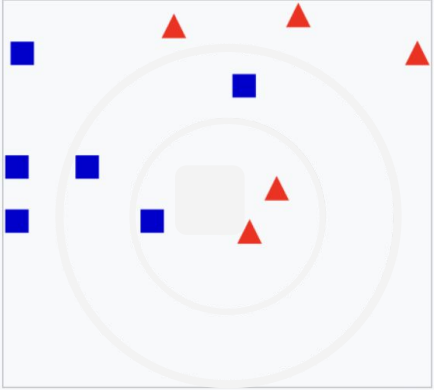
We will explore their advantages and disadvantages as the course progresses.

Our first ML Model

- k-Nearest neighbors (k-NN)

Our first ML Model

- You are given a set of data points of two classes: red triangles, and blue squares

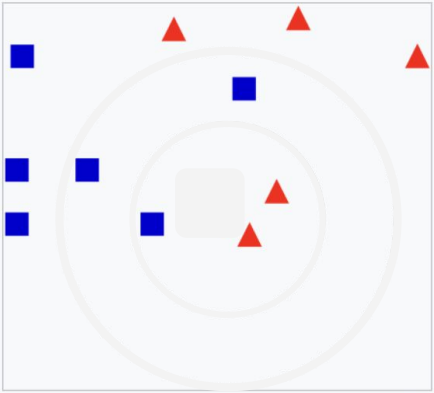


Example of *k*-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

src: https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Our first ML Model

- You are given a set of data points of two classes: red triangles, and blue squares
- And asked to develop a ML model that can classify (a new data point)between these two classes.

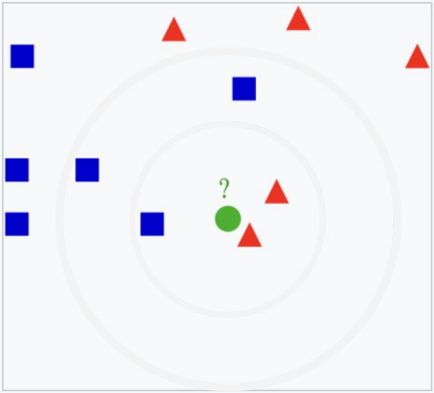


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Our first ML Model

- k-Nearest neighbors (k-NN)
  - Supervised learning
  - Non parametric ***(Distance based method)***
  - Both for Classification and Regression solutions



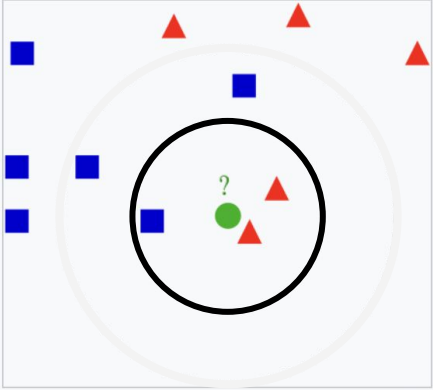
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- For
  - *k = 3, and*
  - *Using L2/Euclidean distance as proximity metric*



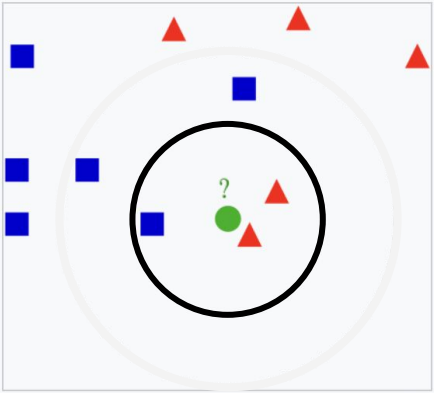
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- For
  - *k = 3, and*
  - *Using L2/Euclidean distance as proximity metric*
  - *The output label would be a Red Triangle:* ▲



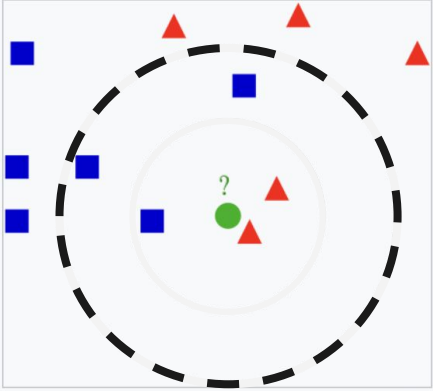
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Our first ML Model

- k-Nearest neighbors (k-NN)
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  - Both for Classification and Regression solutions

- For
  - *k = 5, and*
  - *Using L2/Euclidean distance as proximity metric*
  - *The output label would be a Blue Square:*



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Let's Build Our First Model

kNN Regression

QA